Signed Binary Numbers

We can represent signed binary nos. in 3 formats.

1. Signed Magnitude.
2. Signed 1's complement.
3. Signed 2's complement.

→ In signed magnitude, −6 is obtained from +6 by changing the signbit in leftmost position from 0 to 1.

→ In signed 1's complement, −6 is obtained by complementing all the bits of +6, including the sign bit.

→ In signed 2's complement −6 is obtained by taking 2's complement of positive no. (+6), including sign bit.

0 0 0 0 0 1 1 0 → +6

1 0 0 0 0 1 1 0 → −6

(+6) 1 1 1 1 1 0 0 1 → −6 in 1's comp

1 1 1 1 1 0 1 0 → −6 in 2's comp

−36 :- (8 bit)

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Sign magnitude :-	1	0	0	1	0	0	1	0
1's complement :-	1	1	0	1	1	0	1	1
2's complement :-	1	1	0	1	1	1	0	0

+36

0 0 1 0 0 1 0 0

Binary Addition

$0 + 0 = 0$

$0 + 1 = 1$

$1 + 0 = 1$

$1 + 1 = (10)_2$

```
  1 1 0 0 1 1 0 0
  1 1 0 1 1 0 1 0
-----------------
1 1 0 1 0 0 1 1 0
-----------------
```

Binary Subtraction

$0 - 0 = 0$

$0 - 1 = 1$ (with 1 borrow)

$1 - 0 = 1$

$1 - 1 = 0$

```
          10
        0 Ø  10  0 10
    1   X X  Ø 1 X Ø 0
    0 0 1 1  0 0 1 0
  -------------------
    1 0 1 1  1 0 1 0
  -------------------
```

Binary Subtraction Using 2's Comp

$(28) - (15)_{10}$

$(15)_{10} - (28)_{10}$

Perform $(-42) - (-13) = (-42) + 13$.

$+42 = 0\ 1\ 0\ 1\ 0\ 1\ 0$

$-42 = 1\ 0\ 1\ 0\ 1\ 1\ 0$ (2's comp.

$+13 = 0\ 0\ 0\ 1\ 1\ 0\ 1$

$-13 = 1\ 1\ 1\ 0\ 0\ 1\ 1$ (2's comp).

$$\begin{array}{r} -42 \\ +13 \\ \hline 29 \end{array}$$

$$\begin{array}{r} 0\ 0\ 0\ 1\ 1\ 0\ 1 \\ +(-42)\ 1\ 0\ 1\ 0\ 1\ 1\ 0 \\ \hline 1\ 1\ 0\ 0\ 0\ 1\ 1 \end{array}$$

(No carry is generated the result is -ve.

To verify result take 2's comp of ans $(1\ 1\ 0\ 0\ 0\ 1\ 1)$

$= 0\ 0\ 1\ 1\ 1\ 0\ 1 = 16+8+4+1$

$= 29.$ ✓

Q. (a) $+8 + 12 \rightarrow (10100)_2$

(b) $-8 + 12 \rightarrow (100)_2$

(c) $+8 - 12 \rightarrow (11100)_2$

(d) $-8 - 12 \rightarrow (101100)_2$

OCTAL ADDITION

(1) The sum of two octal digits is the same as their decimal sum, provided the decimal sum is less than 8. If the decimal sum is 8 or greater, subtract 8 to obtain the octal digit.

(2) A carry of 1 is produced when the decimal sum is corrected in above manner.

(a) $(167)_8 + (325)_8$

$$\begin{array}{cccc} & 1 & 1 & \\ & 1 & 6 & 7 \\ + & 3 & 2 & 5 \\ \hline & 5 & 9-8 & 12-8 \\ \hline & 5 & 1 & 4 \end{array}$$

(b) Add the octal nos. $(341)_8$, $(125)_8$, $(472)_8$ and $(577)_8$

$$\begin{array}{cccc} & 2 & 1 & \\ & 3 & 4 & 1 \\ & 1 & 2 & 5 \\ & 4 & 7 & 2 \\ +1 & 5 & 7 & 7 \\ \hline & 15-8 & 21-16 & 15-8 \end{array}$$

$$(1757)_8$$

Octal Subtraction using 8's comp

* Octal Subtraction is performed using 8's complement.

Find 8's comp. of $(346)_8$

7's comp :-

$$\begin{array}{ccc} 7 & 7 & 7 \\ -3 & 4 & 6 \\ \hline (4 & 3 & 1)_8 \end{array}$$

8's comp :-

$$\begin{array}{ccc} 4 & 3 & 1 \\ + & & 1 \\ \hline (4 & 3 & 2)_8 \end{array}$$

Step-1: find 8's comp. of subtrahend.

Step-2: Add two octal nos. (first no. and 8's comp. of the second no.)

Step-3: If carry is produced in the addition it is ignored, otherwise find 8's comp. of the sum as a result with -ve sign.

Q. $(516)_8 - (413)_8$

7's comp of 413 = $\begin{array}{r} 7\ 7\ 7 \\ 4\ 1\ 3 \\ \hline (3\ 6\ 4)_8 \end{array}$

8's comp of 413 = $\begin{array}{r} 364 \\ +\ 1 \\ \hline 365 \end{array}$

$$\begin{array}{r} {\scriptstyle 1\ 1} \\ 516 \\ +1\ 365 \\ \hline (9-8)\,(8-8)\,(11-8) \end{array}$$

(1) 1 0 3 ⇒ Ans. $(103)_8$.

ignore carry.

Q. $(316)_8 - (451)_8$.

7's comp of 451 = $\begin{array}{r} 7\ 7\ 7 \\ -\ 4\ 5\ 1 \\ \hline 3\ 2\ 6 \end{array}$

8's comp of 451 = 327

$$\begin{array}{r} {\scriptstyle 1} \\ 3\ 1\ 6 \\ +\ 3\ 2\ 7 \\ \hline 6\ 4\ \ 13-8 \end{array}$$

= 645 No carry, hence take 8's comp ⇒ $\begin{array}{r} 7\ 7\ 7 \\ -6\ 4\ 5 \\ \hline 1\ 3\ 2 \\ +1 \\ \hline 1\ 3\ 3 \end{array}$

Ans $(-133)_8$

Q. $(456)_8$ and $(341)_8 \rightarrow (1017)_8$

Q Perform $(763)_8 - (101)_8 \rightarrow (662)_8$

Q Add the following $(77)_8 + (107)_8 \rightarrow (206)_8$

HEXADECIMAL ADDITION

→ The sum of two hexadecimal digits is the same as their equivalent decimal sum provided the decimal equivalent is less than 16.

→ If the decimal sum is 16 or greater, subtract 16 to obtain the hexadecimal digit.

→ A carry of 1 is produced when the decimal sum is corrected this way.

⇒ $(3)_{16} + (9)_{16} = (C)_{16}$

⇒ $(9)_{16} + (7)_{16} = (16-16) = 0$ Carry 1 $= (10)_{16}$

⇒ $A_{16} + 8_{16} = 18-16 = 2$ Carry 1 $= (12)_{16}$

Q. Add. $(3\ F\ 8)_{16}$ and $(5\ B\ 3)_{16}$

1		
3	F	8
5	B	3
9	(26-16)	B

$(9\ A\ B)_{16}$.

Q.

```
  1  2   2
     8   9   2
     5   8   F
     1   7   8
+    4   8   E
-------------------
  1 (20-16)(34-32)(39-32)
-------------------
```

$(1\ 4\ 2\ 7)_{16}$.

Hexadecimal Subtraction Using 16's complement.

$(A8C)_{16}$

15's comp → 15 15 15 − A 8 C = 5 7 3

+ 1

16's comp = 5 7 4

Step-1 find 16's comp of subtrahend

Step-2: Add two hexadecimal nos.

Step-3: If carry is produced in the addition, it is discarded otherwise 16's comp. of the sum as a result with −ve sign.

$(CB2)_{16} - (972)_{16}$

16's comp →

```
  15  15  15
- 9   7   2
-------------
  6   8   D
+           1
-------------
  6   8   E
-------------
```

```
    1   1
    C   B   2
1   6   8   E
-------------
(19-16)(20-16)(16-16)
-------------
```

(1) 3 4 0

(1) discarded

Ans. $(340)_{16}$

Q. Use 16's comp method to compute $(3B7)_{16} - (854)_{16}$.

Q. Perform addition of $(FA6)_{16}$ and $(BC4)_{16}$ Ans. $(1B6A)_{16}$

$(E8D)_{16} - (1F8)_{16}$ Ans $(C95)_{16}$

$(1A6)_{16} + (474)_{8}$ Ans. $(2E2)_{16}$

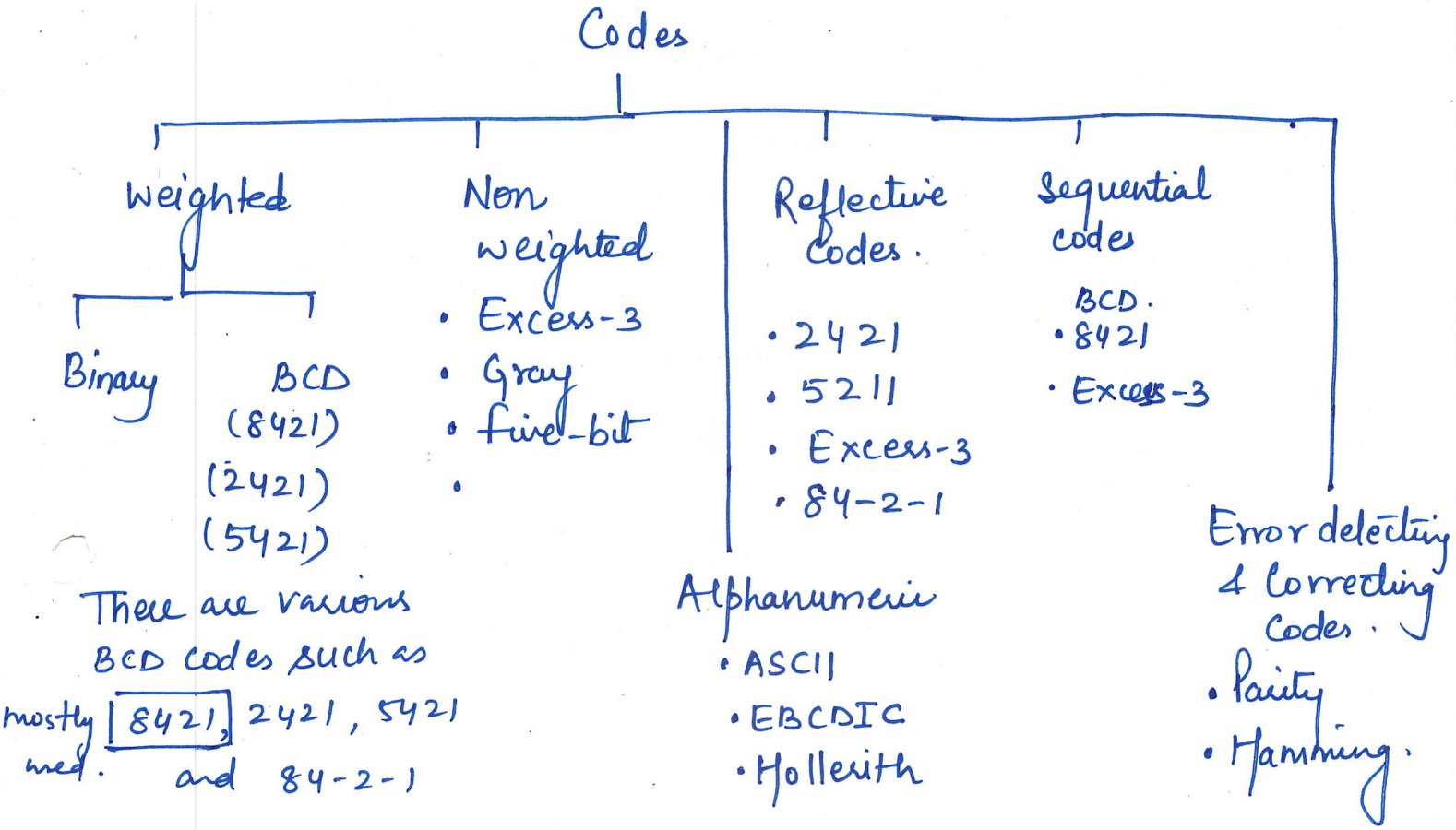
$(E8D)_{16} - (10101010)_{2}$ Ans $(DE3)_{16}$

$(ABC)_{16} + (CDE)_{16}$ Ans $(179A)_{16}$.

BINARY CODES

* When nos., alphabets or words are represented by a specific group of symbols, we say that they are encoded.
* The group of symbols used to encode them are called codes.

Classification of Binary Codes

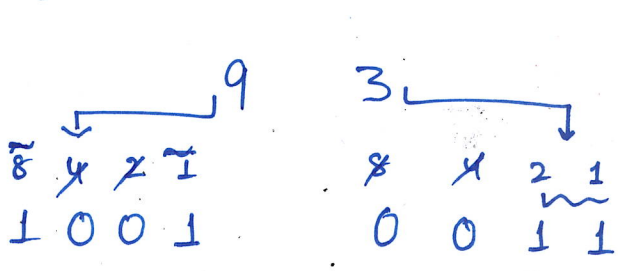


WEIGHTED CODES

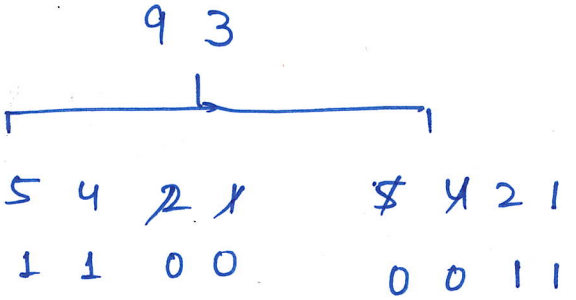
In weighted codes, each digit position of the no. represents a specific weight.

(a) BCD (8421)

In weighted binary code each bit has a weight 8, 4, 2 or 1 and each decimal digit is represented by a group of four bits.



8421 code



5421 code.

( )$_b$ → Decimal → BCD.

→ BCD stands for Binary coded decimal.

→ Simply replace the decimal digit by 4 bit binary equivalent to get BCD Value.

BCD Range (0 to 9) i.e. (0000 to 1001)

$(5C)_{16} \rightarrow (?)_{BCD}$

126.9 → 0001 0010 0110 1001

$(5C)_{16} = 5\times16 + 12\times16^0$
$= (92)_{16}$

$(126.9)_{16} = (0001\ 0010\ 0110\ 1001)_{10} = (1001\ 0010)_{BCD}$

$(621)_8 \rightarrow (401)_{10}$
$\rightarrow (0100\ 0000\ 0001)_{BCD}$

Excess-3 Code (XS3)

Excess-3 Code can be obtained from BCD.

Simply add 3(0011) to the BCD equivalent

$(53)_{10} = (?)_{XS3}$.

$(53)_{16} =$ 0101 0011
+ 0011 0011 (Add 3 to each digit)
1000 0110

$(53)_{16} = (10000110)_{XS3}$.

Excess-3 is sequential code, we will get the next code by adding 1 to it. It is also self complementing code.

0 → 0011
1 → 0100
2 → 0101
3 → 0110
4 → 0111
5 → 1000
6 → 1001
7 → 1010
8 → 1011
9 → 1100

(C) 8 4-2-1 and 2 4 2 1 codes.

MSB →

2421 BCD	5421 BCD	84-2-1 BCD.
0 → 0000	0 → 0000	0 → 0000
1 → 0001	1 → 0001	1 → 0111
2 → 0010	2 → 0010	2 → 0110
3 → 0011	3 → 0011	3 → 0101
4 → 0100	4 → 0100	4 → 0100
5 → 0101	5 → 1000	5 → 1011
6 → 1100	6 → 1001	6 → 1010
7 → 1101	7 → 1010	7 → 1001
8 → 1110	8 → 1011	8 → 1000
9 → 1111	9 → 1100	9 → 1111

8421, 2421, 5421 are +ve weighted codes.

* 84-2-1 are -ve weighted codes and this also self complementing code / Reflective code.

* A self complementing code is a code whose 9's complement can be found just by replacing 1 → 0 and 0 by 1.

9's comp of 1 ⇒ 9 - 1 = 8.

Code of 1 → 0111

Code of 8 → 1000

9's comp of 2 ⇒ 9 - 2 = 7

Code of 2 → 0110

Code of 7 → 1001